

Economics S5040: Dynamic Stochastic Modeling

Lecture 10 — Computing Reservation Wages

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1 Review: Stop or Continue?

The simple McCall model has the Bellman equation

$$V(w) = \max \left\{ \frac{w}{1-\beta}, c + \beta E[V(W')] \right\}.$$

Define the continuation value

$$C = c + \beta E[V(W')].$$

Then

$$V(w) = \max \left\{ \frac{w}{1-\beta}, C \right\}.$$

The reservation wage is defined by indifference:

$$\frac{\bar{w}}{1-\beta} = C.$$

Hence

$$\bar{w} = (1-\beta)C.$$

The policy is

$$\sigma(w) = \begin{cases} 0, & w < \bar{w}, \\ 1, & w \geq \bar{w}, \end{cases}$$

where 1 means Accept and 0 means Reject.

2 A Continuous State Space

Economically, wage offers can be drawn from a continuous distribution

$$W \sim F.$$

Then the state space is a continuum.

The continuation value is

$$C = c + \beta \int V(w') dF(w').$$

This is the continuous-state version of an expectation.

If F has density f , then

$$C = c + \beta \int V(w') f(w') dw'.$$

Thus moving from a finite state space to a continuum changes a sum into an integral, but it does not change the economic logic of the Bellman equation.

3 Why We Often Use a Wage Grid

A computer cannot literally store a value function at every point of a continuum.

One simple computational approach is to choose a grid

$$w_1, \dots, w_n$$

and probabilities

$$q_1, \dots, q_n.$$

Then

$$E[V(W')] \approx \sum_{j=1}^n q_j V(w_j).$$

The continuation value becomes

$$C \approx c + \beta \sum_{j=1}^n q_j V(w_j).$$

We have converted an infinite-state economic model into a finite numerical approximation. It is important to keep these two objects conceptually separate:

economic model: continuous state space
computer approximation: finite wage grid.

4 Value Function Iteration

Suppose the wage grid is

$$\mathbb{W} = \{w_1, \dots, w_n\}.$$

Given a guess V_n , first calculate its implied continuation value:

$$C_n = c + \beta \sum_{j=1}^n q_j V_n(w_j).$$

Then update the value function:

$$V_{n+1}(w_i) = \max \left\{ \frac{w_i}{1 - \beta}, C_n \right\}.$$

Then repeat.

This is exactly the value function iteration from Week 4:

$$V_{n+1} = TV_n.$$

The new feature is that the Bellman operator has an especially simple form.

5 A First Python Implementation

Suppose `w` contains wage values and `q` their probabilities.

```
import numpy as np

def bellman(V, w, q, c, beta):
    continuation = c + beta * (q @ V)
    accept = w / (1 - beta)
    return np.maximum(accept, continuation)
```

The line

`q @ V`

computes

$$\sum_j q_j V(w_j) = E[V(W')].$$

Thus the code mirrors the Bellman equation directly.

6 Iterating to Convergence

We can iterate until successive guesses are close:

```
def solve_mccall(w, q, c, beta,
                tol=1e-10, max_iter=10_000):

    V = np.zeros(len(w))

    for i in range(max_iter):

        V_new = bellman(V, w, q, c, beta)

        if np.max(np.abs(V_new - V)) < tol:
            return V_new

        V = V_new

    raise RuntimeError("Value iteration did not converge")
```

Once V is known, calculate

$$C = c + \beta q'V$$

and then

$$\bar{w} = (1 - \beta)C.$$

7 The Continuation Value as the Heart of the Problem

There is an even simpler way to think about the computation.

Because

$$V(w) = \max \left\{ \frac{w}{1 - \beta}, C \right\},$$

we can substitute this expression directly into the definition of C :

$$C = c + \beta E \left[\max \left\{ \frac{W'}{1 - \beta}, C \right\} \right].$$

Thus the entire problem can be reduced to a scalar fixed-point equation:

$$\boxed{C = c + \beta E \left[\max \left\{ \frac{W'}{1 - \beta}, C \right\} \right].}$$

This is a remarkable simplification.

The original value function is defined over a continuum of possible wages, but the IID structure lets us solve for a single number: the continuation value.

Once C is known,

$$\bar{w} = (1 - \beta)C,$$

and the entire policy follows.

8 Why This Simplification Works

The simplification depends on the IID assumption.

After rejecting an offer, the worker draws again from the same distribution, regardless of today's wage.

Therefore the value of continuing does not depend on w .

If tomorrow's wage distribution depended on today's wage, we would instead have something like

$$C(w) = c + \beta E[V(W') \mid W = w].$$

Then the continuation value itself would vary with the current state.

The simple McCall model is valuable partly because it makes continuation value so transparent.

9 Comparative Statics: Unemployment Compensation

What happens when c rises?

The continuation value is

$$C = c + \beta E[V(W')].$$

A larger c directly raises the payoff from rejecting an offer.

Thus continuing to search becomes more attractive.

The worker becomes more selective, so

$$\boxed{c \uparrow \implies \bar{w} \uparrow.}$$

Higher unemployment compensation raises the reservation wage.

This does not mean that the worker “likes unemployment.” It means that unemployment has become less costly relative to accepting a low wage.

10 Comparative Statics: Patience

Now increase β .

A more patient worker places greater weight on the future benefits of continued search.

Rejecting a mediocre wage today preserves the possibility of receiving a better wage later.

Hence, in the standard McCall model,

$$\boxed{\beta \uparrow \implies \bar{w} \uparrow .}$$

More patient workers are willing to wait for better offers.

This is another example of continuation value doing the economic work.

11 Comparative Statics: Better Wage Opportunities

Suppose the wage-offer distribution improves.

For example, probability shifts from low wages toward high wages.

Then the expected value of continuing to search rises:

$$E[V(W')] \uparrow .$$

Hence

$$C \uparrow$$

and therefore

$$\boxed{\bar{w} \uparrow .}$$

When good future opportunities are more likely, the worker is less willing to settle for a poor offer today.

12 Search Duration

The reservation wage also tells us something about unemployment duration.

Let

$$p = \Pr(W \geq \bar{w})$$

be the probability that a wage draw is acceptable.

With IID offers, each period is another independent chance to draw an acceptable wage.

A higher reservation wage generally lowers p .

Thus policies or parameters that increase $\bar{w}$ tend to increase the expected duration of unemployment.

This connects the worker's dynamic optimization problem to observable labor market outcomes.

13 Adding Job Separation

The baseline McCall model makes a strong assumption:

$$\boxed{\text{Once a job is accepted, it lasts forever.}}$$

Suppose instead that an employed worker loses the job with probability α each period.

Now accepting wage w no longer gives

$$\frac{w}{1 - \beta}.$$

Let $V_e(w)$ denote the value of being employed at wage w and let $V_u(w)$ denote the value of being unemployed with offer w .

The unemployed worker satisfies

$$V_u(w) = \max \{ V_e(w), c + \beta E[V_u(W')] \}.$$

The employed worker satisfies

$$V_e(w) = w + \beta [(1 - \alpha)V_e(w) + \alpha E[V_u(W')]].$$

The logic is unchanged.

The worker still compares

value of accepting

with

continuation value of searching.

But the value of accepting is lower because the job might disappear.

14 Separation and the Reservation Wage

If α rises, jobs become less durable.

Holding other things fixed, the payoff from finding a particularly high wage falls because the worker is less likely to keep that wage for long.

Thus higher job destruction tends to reduce the value of holding out for a high wage.

In the standard model with separation,

$\alpha \uparrow \implies \bar{w} \downarrow.$

The reservation-wage idea survives even though the value of employment is now more complicated.

15 The Larger Lesson

The McCall model is simple, but it illustrates several ideas that recur throughout dynamic economics:

1. The state space can be continuous and infinite.
2. The control space can nevertheless be very small.
3. Current choices are evaluated partly through continuation values.
4. Optimal stopping problems often generate threshold rules.
5. A value function over many states can sometimes be reduced to a low-dimensional fixed-point problem.
6. Comparative statics often work by asking how a parameter changes the value of continuing relative to stopping.

The reservation wage is therefore more than a labor-market result. It is a particularly clean example of how dynamic programming turns a complicated future into a simple decision rule today.

16 What to Remember

The baseline model can be summarized in three equations:

$$V(w) = \max \left\{ \frac{w}{1 - \beta}, C \right\},$$

$$C = c + \beta E[V(W')],$$

and

$$\bar{w} = (1 - \beta)C.$$

These equations tell the whole story:

state $\rightarrow$ continuation value $\rightarrow$ stop/continue comparison $\rightarrow$ reservation wage.